

**NIRMALA MEMORIAL FOUNDATION COLLEGE OF COMMERCE AND SCIENCE
(AUTONOMOUS)**

Re-Accredited by NAAC with B++ Grade & ISO 9001:2015 Certified Recognized under Section 2 (f) & 12 (B) of the UGC Act, 1956.
Kandivali (East), Mumbai-400101.



**IoT-Based Bus Stop System for Public Transport Overcrowding Prediction
Using Machine Learning**

A Project Report

Submitted in partial fulfillment of the requirements For the degree of B.Sc. Information Technology aligned with
the NEP-2020 Community Engagement Project (CEP)

By

Mr. Om Keluskar and Mr. Jashith Agre

Seat No: 24TIT078, 24TIT011

Semester: V

Under the esteemed guidance of

Mr. George Thekkevilayil
Assistant Professor

BACHELOR OF SCIENCE INFORMATION TECHNOLOGY

Academic Year: 2026–2027



Nirmala Memorial Foundation College of Commerce and Science

(Autonomous)

Re-Accredited by NAAC with B++ Grade & ISO 9001:2015 Certified
Recognized under Section 2 (f) & 12 (B) of the UGC Act, 1956

D S Road, Asha Nagar, Thakur Complex, Kandivali (East), Mumbai-400101 Tel: 022-69436400

CERTIFICATE OF PROJECT COMPLETION

B.Sc. (Information Technology)

This is to certify that Mr. Om Keluskar and Mr. Jashith Agre, students of T.Y.B.Sc. (Information Technology), Semester V, bearing Seat No. 24TIT078 and 24TIT011, has successfully completed the Project Work titled "IoT-Based Bus Stop System for Public Transport Overcrowding Prediction Using Machine Learning." The project has been submitted in partial fulfillment of the requirements prescribed in the syllabus for the award of the Degree of Bachelor of Science in Information Technology for the academic year 2026–2027. The practical and project work has been carried out successfully under the supervision of the undersigned.

Internal Examiner

External Examiner

Co-ordinator

Prof. George Thekkevilayil

Prof. Vaishali Mishra

College seal and Date



Nirmala Memorial Foundation College of Commerce and Science

(Autonomous)

Re-Accredited by NAAC with B++ Grade & ISO 9001:2015 Certified
Recognized under Section 2 (f) & 12 (B) of the UGC Act, 1956

D S Road, Asha Nagar, Thakur Complex, Kandivali (East), Mumbai-400101 Tel: 022-69436400

PROJECT PROPOSAL APPROVAL SHEET

**Department of Information Technology
Academic Year 2026-27**

1. Name of the Students: Mr. Om Keluskar and Mr. Jashith Agre

2. Seat Number: 24TIT078, 24TIT011

3. Is this your first submission? Yes No ☐ ☐

4. Name of the Guide: Prof. George Thekkevilayil

5. Teaching Experience of the Guide: 26 Years

The proposed project titled "IoT-Based Bus Stop System for Public Transport Overcrowding Prediction Using Machine Learning." submitted by the above students, has been examined and found suitable for the partial fulfilment of the requirements for Semester V of the B.Sc IT Degree Programme. The project is hereby approved for implementation under the guidance of the undersigned.

Signature of Guide

Date:

Signature of Co-ordinator

Date:

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION AND COMMUNITY CONTEXT	1
1.1. Project Background	1
1.2. Community Problem Identification	2
1.3. Community Need Assessment	3
1.4. Existing System Analysis	4
1.5. Problem Statement	5
1.6. Proposed Solution	5
1.7. Aim and Objectives	6
1.8. Scope of Project	7
1.9. Limitations	8
1.10. Major Modules	8
1.11. Success Criteria	9
1.12. Expected Outcomes	10
1.13. Chapter Summary	10
CHAPTER 2: SYSTEM ANALYSIS AND REQUIREMENT ANALYSIS	11
2.1. Introduction	11
2.2. Literature Review	11
2.3. Comparative Analysis	15
2.4. Research Gap Identification	17
2.5. Requirement Gathering	18
2.6. Stakeholder Analysis	19
2.7. Functional Requirements	20
2.8. Nonfunctional Requirements	21
2.9. Feasibility Study	22
2.10. Software Requirements	23
2.11. Hardware Requirements	24
2.12. Project Schedule	25
2.13. Development Methodology	26
CHAPTER 3: SYSTEM DESIGN	27
3.1. System Architecture	27
3.2. System Workflow	28
3.3. Block Diagram	29
3.4. Functional Flow Diagram	29
3.5. Circuit Diagram	30
3.6. Component Interconnection Diagram	31
3.7. Hardware Module Design	33
3.8. Data Flow & Processing Logic	35
3.9. Database Design (if applicable)	37
3.10. Dashboard/UI Design	38
3.11. Technology Stack	42
3.12. Safety & Security Considerations	43
3.13. Chapter Summary	45

CHAPTER 4: DEVELOPMENT AND IMPLEMENTATION	46
4.1. Development Environment	46
4.2. Implementation Approach	46
4.3. Hardware Assembly	48
4.4. Firmware Development	51
4.5. Cloud / Blynk Integration	56
4.6. Dashboard Development	58
4.7. Module-wise Implementation	59
4.8. Algorithms and Control Logic	60
4.9. Safety Features Implemented	61
4.10. Prototype Development	62
4.11. Prototype Photos & Demonstration	63
CHAPTER 5: TESTING, VALIDATION AND PERFORMANCE EVALUATION	66
5.1. Testing Strategy	66
5.2. Unit Testing	66
5.3. Integration Testing	68
5.4. System Testing	70
5.5. User Acceptance Testing	72
5.6. Performance Evaluation	72
5.7. Security Testing	74
5.8. Validation Metrics	75
5.9. Test Cases	77
5.10. Defects Identified and Corrections	78
5.11. Chapter Summary	79
CHAPTER 6: COMMUNITY DEPLOYMENT AND IMPACT ASSESSMENT	80
6.1. Beneficiary Profile	80
6.2. Community Engagement Activities	80
6.3. Pilot Deployment	81
6.4. User Training	82
6.5. Feedback Collection	83
6.6. Feedback Analysis	83
6.7. Impact Assessment	85
6.8. Outcome Achievement Matrix	86
6.9. Sustainability Plan	86
CHAPTER 7: RESULTS, DISCUSSION, CONCLUSION AND FUTURE SCOPE	87
7.1. Results and Discussion	87
7.2. Achievement of Objectives	88
7.3. Project Contributions	89
7.4. Limitations	90
7.5. Future Enhancements	90
7.6. Conclusion	91

REFERENCES	86
APPENDIX A: SYSTEM LOGIC	89
SURVEY QUESTIONNAIRE	94
FEEDBACK QUESTIONNAIRE	95
FIELD VISIT PHOTOGRAPHS	96
ANNEXURES	94 – 98

Name of Student(s): Mr. Om Keluskar, Mr. Jashith Agre

Seat No: 24TIT078, 24TIT011

Name of Participant: _____

Signature of Participant: _____

Survey Questionnaire

1. How frequently do you experience overcrowding at Dindoshi Bus Depot or the BEST Route 326 bus stop?

☐ Every Day ☐ Several Times a Week ☐ Occasionally ☐ Rarely

2. Which mode of transport do you use most frequently to reach this stop?

☐ BEST Bus (Route 326) ☐ Other BEST Bus ☐ Train then BEST Bus ☐ Walking

3. What is the biggest problem you face because of overcrowding at the bus stop?

☐ Missed bus / cannot board ☐ Pushing or injury risk ☐ Long uncomfortable wait ☐ No crowding warning

4. Have you ever been unable to board a bus because the stop or doorway was too crowded?

☐ Yes, Multiple Times ☐ Yes, Once ☐ No ☐ Not Applicable

5. How safe do you feel while waiting at an overcrowded bus stop?

☐ Very Safe ☐ Safe ☐ Unsafe ☐ Very Unsafe

6. Have you ever reported overcrowding to depot staff or BEST officials?

☐ Yes ☐ No ☐ Didn't Know How ☐ Never Felt It Would Help

7. If you did report overcrowding, how satisfied were you with the response?

☐ Very Satisfied ☐ Satisfied ☐ Dissatisfied ☐ Never Reported

8. How important is it to warn commuters as soon as overcrowding is detected?

☐ Extremely Important ☐ Important ☐ Slightly Important ☐ Not Important

9. Do you think rain, heat, or peak hours make overcrowding at the stop more dangerous?

☐ Strongly Agree ☐ Agree ☐ Disagree ☐ Strongly Disagree

10. Would you trust an automated IoT system that predicts overcrowding using on-site lights, without requiring a mobile app?

☐ Definitely Yes ☐ Probably Yes ☐ Probably No ☐ Definitely No

11. Which feature would be most useful in a smart bus-stop overcrowding system?

☐ On-site LED display ☐ Telegram staff alerts ☐ Live risk percentage ☐ Bus-left operator button

12. Do you believe an IoT-based overcrowding prediction system can improve commuter safety at BEST bus stops?

☐ Strongly Agree ☐ Agree ☐ Disagree ☐ Strongly Disagree

Name of Student(s): Mr. Om Keluskar, Mr. Jashith Agre

Seat No: 24TIT078, 24TIT011

Name of Participant: _____

Signature of Participant: _____

Feedback Questionnaire

1. How easy was it to understand the purpose of the IoT bus-stop overcrowding system?

☐ Very Easy ☐ Easy ☐ Neutral ☐ Difficult

2. How useful do you think automatic passenger counting using the IoT sensor would be for overcrowding awareness?

☐ Very Useful ☐ Useful ☐ Moderately Useful ☐ Not Useful at All

3. How clear and understandable was the demonstrated overcrowding-prediction process?

☐ Very Clear ☐ Clear ☐ Neutral ☐ Unclear

4. How useful were the on-site red and green LED indicators for understanding crowding?

☐ Very Useful ☐ Useful ☐ Moderately Useful ☐ Not Useful at All

5. How useful would real-time Telegram alerts be for depot staff when crowding becomes CRITICAL?

☐ Very Useful ☐ Useful ☐ Moderately Useful ☐ Not Useful at All

6. How easy was it to understand the crowding information shown on the dashboard (count, weather, risk, status)?

☐ Very Easy ☐ Easy ☐ Neutral ☐ Difficult

7. How clearly did the hardware demonstration show how the sensor detects passenger crossings?

☐ Very Clear ☐ Clear ☐ Neutral ☐ Unclear

8. How confident are you that this system can warn of overcrowding faster than manual visual guessing?

☐ Very Confident ☐ Confident ☐ Moderately Confident ☐ Slightly Confident

9. Would you be willing to use this system if it remained available at Dindoshi Route 326?

☐ Definitely Yes ☐ Probably Yes ☐ Not Sure ☐ Definitely No

10. How clear and easy to understand were the NORMAL, WARNING, and CRITICAL status lights?

☐ Very Clear ☐ Clear ☐ Neutral ☐ Unclear

11. How useful do you think this system would be for improving crowding awareness at your bus stop?

☐ Very Useful ☐ Useful ☐ Moderately Useful ☐ Not Useful at All

12. Overall, how satisfied are you with the IoT bus-stop overcrowding system and its potential for commuter safety?

☐ Very Satisfied ☐ Satisfied ☐ Neutral ☐ Dissatisfied